Saying “I’m Not Sure” to Young Learners: A Protocol for Designing and Expert-Reviewing Uncertainty Cues in a Children’s AI Tutor

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Study protocol, version 2 (November 12, 2026). First posted October 8, 2026. No results yet. Not peer reviewed.

What changed in version 2: the study had not started by the original dates, so the schedule now begins November 16, 2026. I also revised the wording and checked every reference against publisher records. The research questions and analysis plan are unchanged. The ethics step now fits my status as an independent researcher (Section 4.3).

Abstract

If an AI tutor should sometimes decline to answer, it also needs a way to say so that a child can understand. Research with adults shows that first-person expressions of uncertainty can reduce overreliance on a language model, and research with children shows that they form views about conversational agents’ intelligence and trustworthiness. How a tutor should express uncertainty to young learners, though, has barely been designed at all. This protocol describes a design study for learners of roughly ages 8 to 11. I will build a scripted, clickable prototype of one short science lesson in five versions that differ only in the uncertainty cue: no cue, a plain-language hedge, a visual confidence meter, a character who shows it is thinking, and an explicit “I don’t know” with a hand-off to a teacher. Children will not see the prototype in this study. Instead, I will evaluate the variants through a heuristic review of my own and walkthroughs with three to five adult experts. The result will be a set of candidate cues and design risks, ready for a later study with children under ethics review.

Keywords: uncertainty communication; children; AI tutors; abstention; trust calibration; heuristic evaluation; prototyping

1. Introduction

In earlier work, I found that a retrieval-augmented question-answering system never declined to answer, even on questions it had no basis for (Choi, 2026). If a tutor *should* sometimes say it does not know, the next question is how it should say it, and children make that question harder. The cue has to be understandable at their reading level. It should point them to a useful next step, such as checking a book or asking a teacher. And it should not make the tutor seem broken or untrustworthy in general.

The question I am working from is this: how can an AI tutor tell a young learner that it is unsure in a way that prompts checking, rather than blind acceptance or blanket distrust? Before involving any child, I want to design candidate answers to that question and get expert criticism of them.

2. Background

Druga et al. (2017) studied how 26 children aged 3 to 10 interacted with four agents, including Amazon Alexa and Google Home, and asked them about trust, intelligence, and personality. They identified four themes, among them perceived intelligence and understanding. What that work leaves open is how agents should signal their own limits to children.

Ruan et al. (2020) worked with 72 children in grades 3 to 5. Embedding a math task in a narrative boosted engagement, and adding a tutoring chatbot (operated by a researcher behind the scenes, in a Wizard-of-Oz setup) improved learning outcomes. The tutor's own uncertainty, however, was not a design variable in that study.

With adults, Kim et al. (2024) found in a pre-registered experiment (N = 404) that first-person uncertainty expressions such as "I'm not sure, but…" decreased agreement with a fictional LLM-infused search engine and increased participants' accuracy. Whether similar cues work for children is untested.

Finally, heuristic evaluation (Nielsen & Molich, 1990) is a way to find interface problems without involving end users, which fits an early design stage like this one. Standard heuristics, however, were not written with children or AI uncertainty in mind, so I add three of my own (Section 4.3).

3. Research Questions

- **RQ1.** What forms can an AI tutor's uncertainty take for young learners, in words, visuals, character behavior, or a hand-off to a person?
- **RQ2.** Which forms do experts judge most likely to be understood by children and to prompt checking rather than blind acceptance?
- **RQ3.** What risks do experts see, such as children losing trust entirely or deciding the tutor is broken?

4. Method

4.1 Design space

Every variant answers the same questions in the same lesson, so the cue is the only thing that changes (Table 1).

Table 1. Uncertainty cue variants.

Variant	Cue	What the child sees or hears
A. No cue (control)	None	A confident answer
B. Plain words	First-person hedge	"I'm not sure about this one, but I think…"
C. Visual meter	Confidence picture	A small gauge icon next to the answer
D. Character	The guide shows it is thinking	The guide character pauses and suggests checking the book
E. Hand-off	Abstain and redirect	"I don't know. Let's ask your

Variant	Cue	What the child sees or hears
		teacher or look it up together.”

4.2 Prototype

The prototype is scripted rather than driven by a live model, so every variant shows identical answers. It covers one short lesson (for example, why leaves change color) with six questions: three the tutor answers correctly, two it answers incorrectly, and one it cannot answer. I will build it either as a clickable Figma prototype or as a single web page with a switcher between variants. The guide character will be an original drawing made for this study.

4.3 Evaluation

Heuristic review. I will score each variant against Nielsen’s heuristics plus three I wrote for this problem: the cue is understandable at the target reading level; the cue invites a next action, such as checking; and the cue does not make the tutor seem broken.

Expert walkthroughs. Three to five adult experts (teachers or HCI researchers) will each click through all variants in a randomized order, for about 30 minutes, and answer the same three questions for each one: would a child understand this, what would the child do next, and what could go wrong.

Ethics review. I am doing this study as an independent researcher and have no institutional review board of my own. Before inviting any experts, I will ask an independent IRB whether the walkthroughs need review, and follow its determination. Each expert will get a short written information sheet, can stop at any time, and will be named in the paper only with their permission. I will record only their comments on the prototype and their professional role. If I cannot get a determination in time, I will drop the walkthroughs, run the heuristic review alone, and report it as a single-author review.

5. Analysis Plan

- **RQ1:** the design space itself, documented with screenshots of each variant.
- **RQ2:** heuristic scores per variant, and expert judgments summarized by variant with counts.
- **RQ3:** expert comments coded by theme. For every theme I report, I will say how many experts raised it.

6. Ethics and Limitations

No children take part in this study. Any later study with children will need ethics review and parental consent.

The limitations are real ones. Adult judgments about children are predictions, not evidence of how children actually respond. A scripted prototype cannot show how cues behave with a real model’s errors. And three to five experts can give direction but not generalizable findings.

7. Timeline

Table 2. Planned schedule (2026–2027). No part of the study had started when this revision was posted.

Stage	Dates
Request IRB determination; script and design the variants	Nov 16 to Nov 29, 2026
Build the prototype	Nov 30 to Dec 13, 2026
Heuristic review	Dec 14 to Dec 20, 2026
Expert walkthroughs (if approved)	Jan 4 to Jan 17, 2027
Writing the paper	Jan 18 to Jan 31, 2027
Posting the paper and prototype	Feb 1, 2027

8. Declarations

Origin of this work. This study extends the abstention problem described in Choi (2026) to the design of uncertainty cues for children.

Use of AI tools. Claude (Anthropic) helped draft the decision, revise the wording.

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Competing interests. None.

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